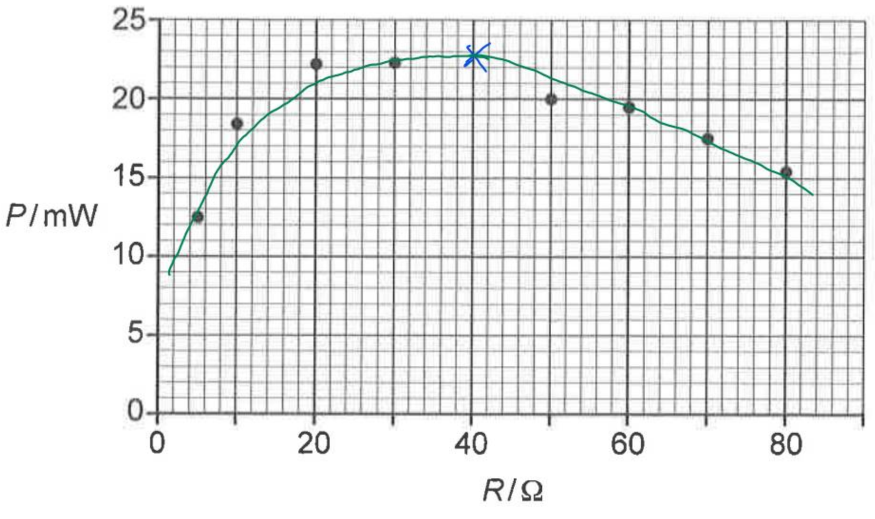
No.	Solution
1(a)(i)	$V = JC^{-1}$ $= kgms^{-2}m(As)^{-1}$ $= kgm^2s^{-3}A^{-1}$
1(a)(ii)	The potential difference (p.d.) between two points in a circuit is the **energy converted from electrical to other forms per unit charge** passing from one point to the other. On the other hand, electromotive force (e.m.f.) of a source is the **energy converted from other forms to electrical per unit charge** driven through the source.
1(b)(i)	$V_R = \frac{100}{100+25}(1.5)$ $= 1.2\text{ V}$
1(b)(ii)	$I = \frac{V_R}{R}$ $= \frac{1.2}{100}$ $= 0.012\text{ A}$
1(c)(i)	$I_{avg} = \frac{18+19+23}{3}$ $= 20\text{ mA}$
1(c)(ii)	$P = (I_{avg})^2 R$ $= (23\times10^{-3})^2(40)$ $= 21.16\text{ mW}$ $\approx 21\text{ mW}\ (2\text{ s.f.})$
1(c)(iii)	

No.	Solution
1(c)(iv)	When *P* is maximum, value of *R* is $25\ \Omega$. Since the internal resistance of the cell is also $25\ \Omega$, both *R* and internal resistance of cell are equal.
1(d)	To have the sound system to be as loud as possible, the power dissipated in the speaker must be the maximum. Based on data from (b) and (c), the sound system should hence be designed such that the resistance of the speaker (i.e. resistance of external load) is equal to the resistance of the amplifier (i.e. internal resistance). <u>Further information (optional):</u> This is also in accordance with the maximum power theorem that to obtain maximum external power from a source with a **finite** internal resistance, the resistance of the external load (i.e. speaker) must be equal to the internal resistance (i.e. amplifier) of the source.
2(a)	Momentum of an object is the product of its mass and its velocity.